Shanghai, China
zhaohaiteng@pku.edu.cn

Haiteng Zhao

Researcher at Shanghai AI Lab

[Personal Website]
[Google Scholar]

AI researcher specializing in advanced reasoning, foundation models, autonomous agents, and AI for science. I am a researcher at Shanghai AI Lab. I received my Ph.D. in Machine Learning and Intelligence from Peking University in 2025. I have hands-on experience contributing to trillion-scale foundation models, reinforcement learning systems, and advanced reasoning agents through research roles at Shanghai AI Lab, Tencent, ByteDance, and Microsoft Research Asia. My research develops scalable learning methods for mathematical reasoning, scientific discovery, and generalizable intelligent agents, with 20+ publications at ICLR, NeurIPS, ICML, ACL, COLM, CVPR, among others.

Education

Ph.D. in Machine Learning and Intelligence, Peking University, advised by Zhihong Deng — **September 2020 – July 2025**
Bachelor of Science in Psychology, Peking University — **September 2016 – July 2020**

Selected Research

★ indicates first or co-first author

Advanced Reasoning & AI for Mathematics

- **Olympiad-Level Mathematical Agents** Developed informal and formal mathematical agents (Intern-S1-MO and InternGeometry) that scored 102 in the 2025 CMO AI track, reaching the Gold Medal threshold and ranking 3rd among all human contestants.
  Achieving Olympia-Level Geometry Large Language Model Agent via Complexity Boosting Reinforcement Learning ★, ICLR 2026
  Long-horizon Reasoning Agent for Olympiad-Level Mathematical Problem Solving, COLM 2026
- **Research-Level Reasoning and Self-Evolution** Proposed inverse-harness training, achieving an 8-point gain on Ph.D.-level math through pure model self-evolution, and built agents to collaborate with experts on research-level mathematical problems.
  Two manuscripts in preparation ★

Foundation Models & Multimodal Pretraining

- **Trillion-Scale Scientific Foundation Models** Contributed advanced mathematical reasoning and multimodal capabilities to the Intern-S model family, including Intern-S1, Intern-S1-Pro, and Intern-S2-Preview.
  Intern-S2-Preview: Scientific Agentic Foundation Model, arXiv 2026
  Intern-S1-Pro: Scientific Multimodal Foundation Model at Trillion Scale, arXiv 2026
- **Scalable Visual Pretraining for Language Intelligence** Explored next-patch prediction methods for pretraining language models on large-scale PDF corpora visually, demonstrating that pure visual pretraining can improve textual language capabilities.
  Exploring Visual Pretraining for Learning Language Intelligence, CVPR 2026
  Scalable Visual Pretraining for Language Intelligence, arXiv 2026

Autonomous Agents & AI for Science

- **Agent Skill Learning and Evaluation** Developed methods that enable language-model agents to create, refine, and reuse action skills during task execution, and contributed to evaluations of multimodal agents in realistic scientific workflows.
  Empowering Large Language Model Agents through Action Learning★, COLM 2024
  ScienceBoard: Evaluating Multimodal Autonomous Agents in Realistic Scientific Workflows, ICLR 2026
- **Multimodal Learning for Biology and Chemistry** Developed multimodal language models for molecular generation, zero-shot molecular learning, protein representation learning, and biological reasoning.
  Selected work:
  GIMLET: A Unified Graph-Text Model for Instruction-Based Molecule Zero-Shot Learning ★, NeurIPS 2023
  Are More Layers Beneficial to Graph Transformers? ★, ICLR 2023

Publication record: 20+ papers and preprints, including publications at ICLR, NeurIPS, ICML, ACL, EMNLP, COLM, and CVPR. Complete list: Personal Website / Google Scholar.

Experience

Researcher — **July 2025 – Present**
Large Model Center — Shanghai AI Lab

Research Intern — **October 2024 – July 2025**
Large Model Center — Shanghai AI Lab

Research Intern — **March 2024 – October 2024**
AI Lab NLP — Tencent

Research Intern | **August 2023 – February 2024**
Data-AML-US, mentored by Jingjing Xu | ByteDance

Research Intern | **September 2022 – July 2025**
HKU NLP, advised by Lingpeng Kong and Qi Liu | The University of Hong Kong

Research Intern | **January 2022 – October 2022**
Natural Language Computing Group, mentored by Shuming Ma | Microsoft Research Asia

Collaborator | **June 2021 – July 2025**
NAIL Group, advised by Luu Anh Tuan | Nanyang Technological University

Research Assistant | **July 2018 – July 2020**
Sigma Lab, Department of Machine Intelligence | Peking University

Awards & Honors

2022 Guorui Scholarship, Peking University
2022 Merit Student Award, Peking University
2018 Scientific Research Award, Peking University